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YOLO26-GE: synergizing lightweight convolution and efficient attention for UAV-based road crack detection

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Abstract

In unmanned aerial vehicle (UAV)-based road crack detection, fine-grained targets and complex backgrounds present a sharp trade-off: lightweight models suffer from insufficient accuracy, whereas high-accuracy models fail to meet real-time requirements. To address this issue, we propose YOLO26-GE, a lightweight detector that synergistically integrates GSConv (ghost shuffle convolution) and efficient multi-scale attention (EMA) into the YOLO26 framework. By replacing standard convolutions with GSConv, we eliminate computational redundancy, reducing the model to only 2.5 million parameters and 5.1 GFLOPs while preserving fine-grained crack texture. We further embed the EMA module to dynamically amplify crack-related feature channels and suppress background noise, thereby enhancing the discriminability of weak-feature targets. Evaluated on the public UAV PDD2023 dataset, YOLO26-GE achieves an inference speed of 714 FPS, with an $F1$ score of 0.724 $\pm$ 0.015, mAP50 of 0.722 $\pm$ 0.008, and mAP@50:95 of 0.415 $\pm$ 0.005. These results substantially surpass the baseline YOLO26 and other models. Together, they demonstrate that YOLO26-GE effectively combines lightweight design with high accuracy in real-time UAV-based road crack detection.

1. Introduction

Pavement cracks are the most prevalent early-stage form of road distress. Without prompt detection and maintenance, they can develop into severe defects such as potholes and subsidence, which compromise traffic safety and substantially increase the road's life-cycle maintenance costs. At the end of 2024, China's total road length reached 5.4904 million km, with expressways accounting for 190 700 km, the largest network worldwide [1]. Given the extensive scale of China's road network, traditional manual inspection can no longer meet the demands of routine, high-frequency inspection. It suffers from low efficiency, inherent subjectivity, and considerable safety risks. In the 'Fourteenth Five-Year Plan for Highway Maintenance Management Development,' the Ministry of Transport explicitly states: digital technologies should drive the upgrading of highway maintenance management, and we should accelerate the development of automated inspection equipment [2].

Unmanned aerial vehicle (UAV)-based imaging offers high efficiency, extensive coverage, and non-contact operation. Progressively supplanting manual and vehicle-based inspection, it has become the dominant paradigm for road crack detection. However, the detection algorithms are dually challenged in accuracy and real-time performance by subtle crack targets and complex, highly variable backgrounds within aerial images.

In recent years, target detection algorithms based on deep learning have developed rapidly and are mainly divided into two categories: two-stage and single-stage. The two-stage method represented by the R-CNN (regions with convolutional neural networks) series uses the process of 'candidate region

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generation-feature extraction-classification regression,' which has high detection accuracy but high computational overhead. The single-stage method represented by the SSD (single shot multibox detector) and YOLO (You Only Look Once) series achieves end-to-end prediction and has obvious advantages in real-time performance. Among them, the YOLO series has become the mainstream algorithm for UAV road crack detection due to its fast detection speed, high accuracy and flexible deployment. However, existing methods still face the dilemma of insufficient accuracy of lightweight models and slow inference speed of high-precision models, making it difficult to meet the detection performance and real-time requirements of drone inspections at the same time. And it can be mainly divided into three different technical branches: lightweight convolution optimization scheme, attention-enhanced detection network, and small target detection model for drones.

Driven by the rapid evolution of single-stage YOLO detectors, numerous improved variants have been developed for pavement crack detection. Existing UAV crack detection methods fall into three technical branches, which are reviewed as follows.

First, lightweight convolution schemes reduce computation for edge hardware yet sacrifice fine crack texture. Tang *et al* [3] and Zhu *et al* [4] developed YOLO variants for embedded real-time detection, while Ren *et al* [5] and Zhu and Xie [6] optimized mobile deployment for road inspection sensors. Beyond YOLO-based detectors, pixel-level lightweight algorithms also exist: Kusumaningrum *et al* [7] adopted K-NN for damage grading, Mohamed *et al* [8] used beamlet transform for segmentation, and Kang *et al* [9] achieved 93% measurement accuracy via distance transform. Meanwhile, Krizhevsky *et al* [10], Omoebamije *et al* [11], and Sanchez *et al* [12] built lightweight CNNs or simplified zYOLOv5 to handle aerial or noisy asphalt images. Despite their computational efficiency, all these methods undergo irreversible texture loss under aggressive compression, leading to frequent missed detections of ultra-fine aerial cracks.

Second, attention-enhanced networks strengthen weak crack features but introduce extra computational overhead without matching lightweight efficiency. Li and Wang [13] incorporated hybrid attention into YOLOv8-RD, yet it still performs poorly on extremely thin cracks. Sohaib *et al* [14] benchmarked multiple attention-based YOLO models, and Pham and Nguyen [15] integrated an attention-augmented YOLOv5 into a monitoring platform. Although these approaches improve feature discrimination, their added modules violate the low-power, real-time constraints of typical UAV processors.

Third, UAV-specific tiny-crack models optimize multi-scale and coordinate mapping but lack a joint lightweight-attention optimization strategy. Ding *et al* [16] designed SCD-YOLO with a shift cross-scale dynamic head for multi-scale road cracks. Ma *et al* [17] combined PCGAN with YOLO-MF and median flow for embedded UAV systems. Jawaharlalnehru *et al* [18] upgraded YOLOv2 with anchor redesign and coordinate transformation modules to link detections with UAV geolocation. Similarly, Wang *et al* [19] and Qui and Liu [20] separately enhanced YOLOv8 and YOLOv3 for aerial small targets. Collectively, these works conduct unilateral small-object optimization, but none of them synergistically compress computational cost while simultaneously compensating for faint crack features in a unified framework.

Systematic review of existing literature reveals two prominent research gaps. Most prior methods only implement one-sided optimization: lightweight convolution compression inevitably destroys crack texture information, while independent attention modules introduce excessive computational overhead, and few dedicated frameworks are designed for low-texture elongated cracks captured by UAVs. Furthermore, current approaches merely stack lightweight units and attention blocks without constructing the bidirectional coupling mechanism of 'shallow compression, high-level compensation,' failing to balance edge deployment constraints and detection precision for subtle pavement cracks.

YOLO26 n, a newly released lightweight YOLO model, is purpose-built for edge deployment, achieving competitive accuracy with fewer parameters and lower GFLOPs. It features an NMS-free inference pipeline and CPU-optimized computation, making it inherently suitable for UAV detection tasks. Nevertheless, it still cannot simultaneously meet the dual demands of lightweight inference and high-precision crack recognition in aerial inspection tasks. The inherent dilemma originates from two practical challenges: aerial cracks present slender shapes, weak textures and low contrast, and are easily confused with shadows and lane markings, triggering frequent missed and false detections; meanwhile, conventional lightweight operations lead to irreversible feature loss during compression, further weakening the model's ability to identify ultra-fine cracks.

In summary, current UAV-based road crack detection research faces three major challenges:

(1) Inherent trade-off in detection frameworks. Standard convolutional (SC) networks incur high computational costs, hindering real-time deployment on UAV edge devices. Lightweight convolution reduces inference cost but severely degrades the weak texture features of fine cracks, causing substantial accuracy drops.

(2) Lack of attention-based feature compensation. Most lightweight methods merely compress the model via convolution structure without an efficient attention mechanism. Without attention guidance, these networks fail to selectively enhance weak crack features or suppress background clutter—such as road shadows and lane markings—leading to high miss and false-detection rates.

(3) Simple module stacking without synergistic co-optimization. Current approaches often stack lightweight convolution and attention modules naively, lacking a bidirectional co-optimization framework. This fails to resolve the fundamental bottleneck of balancing computational cost and detection accuracy in UAV crack inspection.

To address these gaps, we adopt YOLO26 as the baseline and propose YOLO26-GE, a framework that integrates ghost shuffle convolution (GSConv) and efficient multi-scale attention (EMA) to jointly optimize model compactness and detection performance.

2. Method

2.1. YOLO26

YOLO26 n is a representative lightweight object detector. Its unique structural and performance advantages make it particularly suitable for UAV-based road crack inspection and edge deployment, compared to mainstream variants such as YOLOv8n and YOLO11n.

End-to-end deployment. YOLOv8 and YOLO11 still rely on separate NMS post-processing and a DFL branch. On low-power UAV CPUs, the NMS sorting step introduces unstable inference latency. Moreover, the intensive softmax operations in the DFL module often cause precision degradation during INT8 quantization and model export to ONNX/TensorRT. This raises compatibility risks for edge computing boards. In contrast, YOLO26 removes the DFL branch entirely and uses native one-to-one label matching to achieve NMS-free, end-to-end inference. This design eliminates all extra post-processing pipelines, stabilizes inference latency, and substantially lowers the deployment barrier on drone-embedded hardware.

CPU real-time inference. At a unified input resolution of 640 $\times$ 640, YOLOv8n has a CPU inference latency of 80.4 ms, YOLO11n takes 56.1 ms, and YOLO26 n requires only 38.9 ms. This represents a 43% speedup without any increase in parameters or GFLOPs. For GPU-less UAV platforms, this speed gain ensures stable, real-time detection above 30 FPS during extended aerial patrols [21].

Detection of weak, tiny cracks. Pavement cracks in UAV images are low-contrast, elongated micro-targets that occupy very few pixels. YOLOv8 and YOLO11 use fixed SPPF pooling and conventional label assignment. These easily lose fine crack textures after multi-layer downsampling, leading to high miss rates. YOLO26 employs (small-target-aware label matching) and ProgLoss (dynamic loss balancing). During training, these mechanisms increase the feature weight of faint, narrow cracks, reducing the tiny-target miss rate from 11.7% (YOLO11n) to 5.2%. This performance is well-suited to the weak-texture characteristics of cracks in aerial images [21].

The benchmark results in table 5 confirm that vanilla YOLO26n achieves superior mAP@0.5 and mAP@0.5:0.95 with a comparable lightweight parameter count. For these four reasons, we select YOLO26n as the baseline detection framework in this study.

It has a modest parameter count, high computational efficiency, and deployment convenience. As a result, it becomes a preferred model for object detection tasks in resource-constrained scenarios, such as those involving UAVs. The network's core design principle maximizes feature extraction capacity while simplifying the network architecture and reducing computational complexity. This design meets the real-time inference demands of edge devices. Accordingly, we adopt YOLO26 as the baseline model for the improvement study in this work.

YOLO26 adopts the standard Backbone–Neck–Headed three-stage detection framework and provides multiple scale variants. In this paper, YOLO26n is selected as the baseline due to its shallow network, fast inference speed and balanced accuracy. Its overall architecture is shown in figure 1.

In the original YOLO26 backbone, SC operations are utilized, which incurs considerable computational overhead; additionally, the model is not optimized for the elongated and weakly textured characteristics of cracks. Consequently, feature loss is readily introduced during lightweight deployment, thereby impairing detection accuracy. Furthermore, the native attention module is confined to the terminal stage of the backbone, by which the enhancement of high-level global semantic features is constrained.

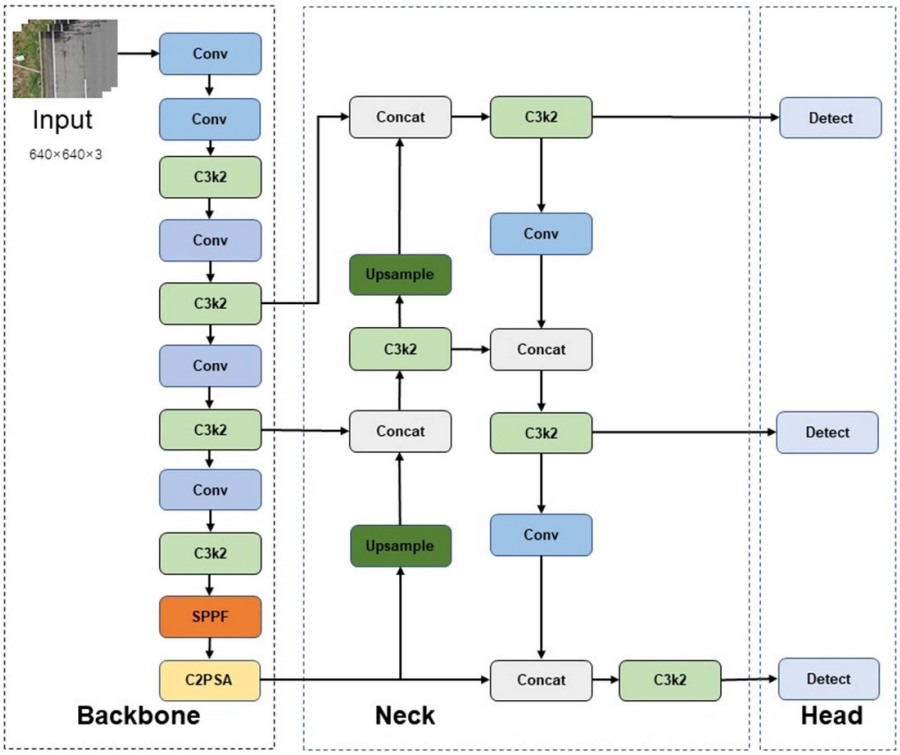


Figure 1. The network structure of the YOLO26 model.

When the model encounters pavement background clutter and illumination interference, it frequently misses fine cracks and generates false detections in complex regions.

We propose YOLO26-GE, a framework that integrates GSConv and EMA for collaborative optimization. Its design is motivated by the deployment potential of YOLO26 and the limitations of current UAV crack detection methods.

First, to balance computational cost and feature retention, we replace shallow SC with GSConv. Its mixed convolution and channel shuffle reduce computations while preserving fine-crack texture details, mitigating the feature degradation common in lightweight models.

Second, to compensate for the lack of attention, we incorporate an efficient EMA mechanism. GSConv alone cannot fully restore features weakened by compression. EMA performs adaptive calibration on high-level fused features, strengthening low-contrast cracks and suppressing background noise. This significantly lowers miss and false-detection rates for fine cracks.

Finally, we design a coupled architecture: GSConv for shallow compression and EMA for high-level compensation. GSConv aggressively lightens the model and frees computational resources, while EMA uses the surplus capacity for precise feature enhancement. Together, they form a mutually reinforcing system that achieves a dual balance—real-time inference and high-precision recognition for UAV crack detection.

2.2. GSConv lightweight convolutional module

Traditional depthwise separable convolution (DSC) substantially reduces the parameter count and computational load. However, it severs inter channel information interaction. This leads to a pronounced degradation in feature representation capability.

In contrast, SC preserves complete channel correlations. But it entails a high computational cost.

To strike an optimal balance between model compactness and feature representation capability, we introduce the GSConv module in this work.

2.2.1. The core principle of GSConv

We adopt GSConv, a lightweight convolution designed for real-time detection [22]. It reduces computational load via grouped convolution, and introduces a channel shuffle mechanism to mitigate feature fragmentation caused by the grouping. The structure of GSConv is illustrated in figure 2.

We define SC as a standard 2D convolutional layer. We configure it with a 2D batch normalization layer and an activation layer.

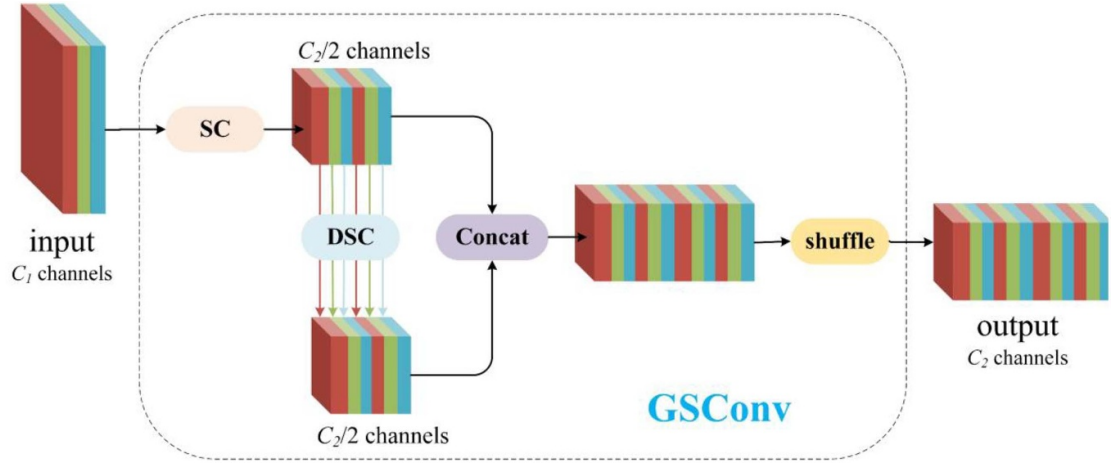


Figure 2. Schematic diagram of the GSConv structure.

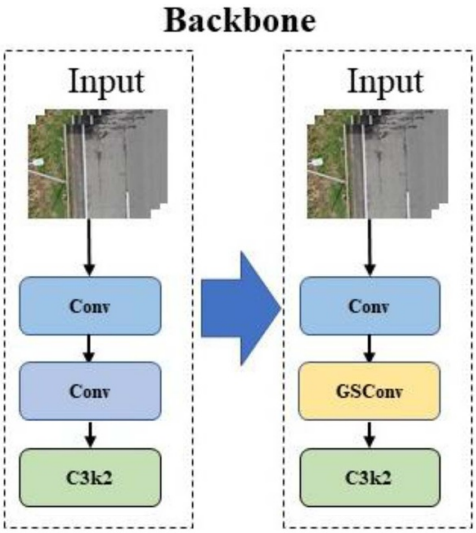


Figure 3. Improvement and deployment of GSConv in YOLO26.

We define DSC as a depthwise separable 2D convolutional layer. We also configure it with a 2D batch normalization layer and an activation layer.

The core principle is the synergistic complementarity between SC and depthwise convolution.

We divide the features into two paths. In one path, we use SC to model global inter-channel dependencies. This ensures feature representation capacity. In the other path, we use depthwise convolution to perform spatial filtering. This substantially reduces the parameter count and computational load.

Then we concatenate the feature maps from the two paths. We execute channel shuffling after concatenation. Channel shuffling realizes uniform information permeation and global fusion.

As a result, the lightweight structure attains feature representation performance comparable to that of SC.

2.2.2. Improvement and deployment of GSConv in YOLO26

To reduce the computational load of the model while preventing an excessive degradation of feature extraction capacity caused by lightweight operations, GSConv is embedded into the shallow feature extraction stage of the YOLO26 backbone, where the original SC layers are replaced, as illustrated in figure 3.

Specifically, we completely replace the standard downsampling convolution at the first layer (P2/4 stage) of the YOLO26 Backbone with GSConv. We keep the output channel count (128), the downsampling stride (2), and the receptive field unchanged. We only optimize the internal computational logic of the convolution. We do not modify the overall network hierarchy.

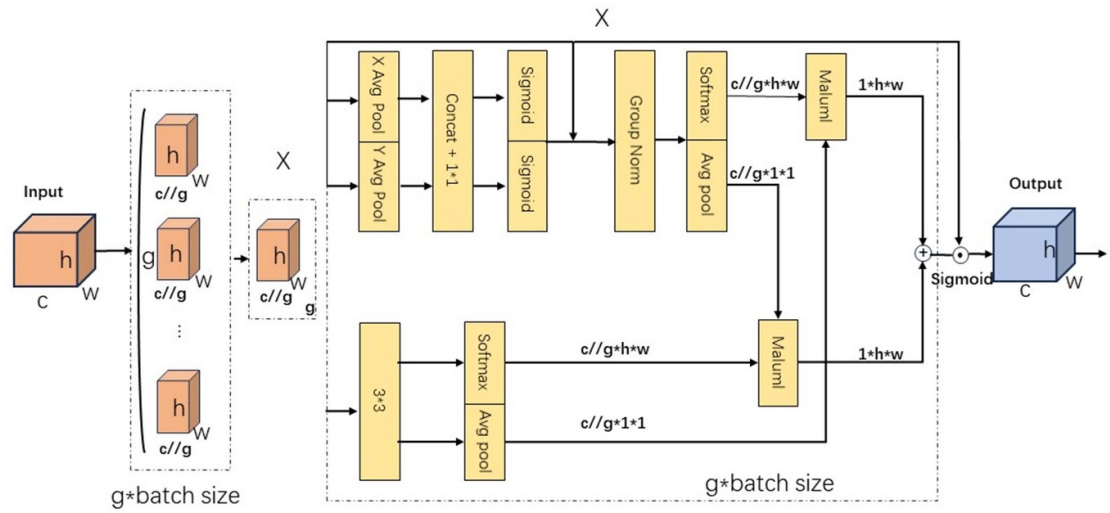


Figure 4. EMA structure diagram.

In the original model, this layer uses a standard 3 × 3 convolution for initial downsampling and feature extraction. Therefore, we identify it as a critical node for lightweight optimization

The above deployment strategy relies on the following considerations.

First, shallow features mainly consist of low-level information, such as textures and edges. These features have limited reliance on sophisticated feature extraction capacity. Therefore, replacing SC with GSConv does not cause significant performance degradation.

Second, shallow feature maps have a large spatial size. The computational cost of SC scales quadratically with the feature map dimensions. Thus, GSConv's lightweight nature effectively reduces the computational burden at this stage. This reserves computational resources for subsequent deep feature extraction and neck fusion.

Third, GSConv has a straightforward architecture. It requires no additional inference optimization tools. We can seamlessly integrate it into the end-to-end inference pipeline of YOLO26 without losing deployment convenience.

2.3. The principle and improved design of EMA attention mechanism

Mainstream attention mechanisms, such as SE, CBAM, and CA, commonly rely on channel dimensionality reduction to constrain computational load. However, this reduction operation loses partial channel information. As a result, it impairs deep feature extraction.

To address this limitation, we introduce the EMA module in this work. The EMA module realizes efficient multi-scale feature enhancement without dimensionality reduction.

2.3.1. The core principle of the EMA attention mechanism

EMA is introduced as a lightweight multi-scale attention mechanism, where channel dependencies and spatial multi-scale information are jointly modeled at a negligible parameter overhead [23]. Its architecture is illustrated in figure 4.

Here, $^{*}g^{*}$ denotes the number of groups. *X* Avg Pool and *Y* Avg Pool represent 1D horizontal and vertical global average pooling, respectively.

Unlike conventional channel attention, EMA does not rely on fully connected layers or complex dimensionality reduction operations. Instead, it divides the input feature channels into multiple groups. It then extracts multi-scale representations along two parallel branches: 1 × 1 convolution and 3 × 3 convolution. After that, it aligns the feature weights through cross-scale interaction.

This mechanism dynamically amplifies the feature responses of fine-textured and small targets such as cracks. It also suppresses background interference from lane markings, shadows, and the like. Therefore, EMA is theoretically better suited for the 'weak-feature, complex-background' scenario in UAV-based road crack detection.

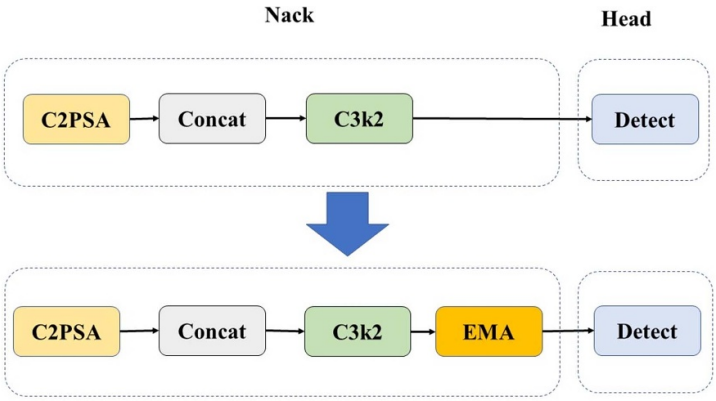


Figure 5. Improvement and deployment of EMA attention mechanism in YOLO26.

2.3.2. Improvement and deployment of EMA attention mechanism in YOLO26

To enhance the model's focusing capability on critical features without substantially increasing computational overhead, the EMA attention mechanism is embedded into the Neck stage of YOLO26, where it operates on deep high-dimensional feature outputs, as illustrated in figure 5.

Specifically, we embed the EMA attention module at the Neck output of YOLO26. This enhances the capacity of high-level features for representing road crack targets.

In the original workflow, the features output by C2PSA first go through concatenation and the C3k2 module. Then they go directly to the detect head. In our improved design, we insert an EMA module after the C3k2 module. The EMA module applies adaptive channel-wise and spatial weighting to the fused high-level features. This enhances crack target features and suppresses complex background interference.

This deployment relies on the following three considerations.

(1) Feature dimension compatibility: EMA has a stronger multi-scale modeling capacity for high-dimensional features. The high channel dimensionality allows us to fully leverage the advantages of its grouped-parallel branch structure.

(2) Lightweight complementarity: the extra parameter overhead and computational cost introduced by EMA are negligible. EMA forms a synergistic optimization with the shallow compression achieved by GSConv. We describe this synergy as 'lightweight early, precise late.' Thus, EMA preserves the overall real-time inference performance.

(3) Detection accuracy optimization: by recalibrating high-level semantics with attention, the model with EMA specifically reduces false negatives (FN) and false positives (FP) caused by the cracks' weak textures.

2.4. The collaborative integration mechanism of GSConv and EMA attention mechanism

GSConv reduces the computational load of YOLO26. However, it also inevitably suppresses certain feature responses. For weakly textured targets like cracks, we observe a relative decline in feature intensity.

Furthermore, background interference in aerial images—such as lane markings and shadows—further exacerbates the semantic confusion between cracks and the background. This confusion leads to missed and false detections.

To address this issue, we introduce the EMA attention mechanism and form a synergistic optimization framework with GSConv. The integration of the two modules is not a mere stacking of components. Instead, we achieve deep coupling at two levels: computational resource allocation and feature information flow.

(1) Computational resource synergy

GSConv significantly reduces the model parameters and computational load. This reduction reserves sufficient inference budget for embedding EMA. In the complete model, the total computational cost and

parameter count remain lower than those of the original YOLO26. Thus, the model ensures real-time performance on edge devices.

(2)Feature quality synergy
GSConv uses channel shuffling to preserve the fine texture structures of cracks. This supplies more informative input features to EMA. On this basis, EMA dynamically enhances crack responses and suppresses background noise along both channel and spatial dimensions. This realizes a 'compensatory refinement' of the GSConv output features.

(3) Accuracy–lightweight balance
Introducing GSConv alone reduces computational overhead, yet inevitably leads to accuracy degradation owing to information fragmentation caused by group convolution. Once we embed the EMA module, it effectively compensates this feature loss via feature recalibration. As a result, the final model achieves competitive detection accuracy while maintaining lightweight characteristics.

From the perspective of information flow, the synergy between GSConv and EMA can be formally described as follows. Let the input feature map be denoted by $\boldsymbol{X}\in\mathbb{R}^{C\times H\times W}$. The output of GSConv is expressed as $\boldsymbol{X}_{\mathrm{G}}=f_{\mathrm{GS}}(\boldsymbol{X})$. Uniform mixing of inter-channel information is realized via channel shuffle, which preserves the directionality and continuity of crack textures. Subsequently, X_{G} propagates through the backbone and neck and is fed into the EMA module. Within EMA, channels are partitioned into g groups, and multi-scale features are extracted through two parallel branches with 1×1 and 3×3 convolutions, respectively. The resulting attention weight matrix is obtained as $\boldsymbol{A}=\sigma\left(f_{1\times 1}(\boldsymbol{X}_{\mathrm{G}})\odot f_{3\times 3}(\boldsymbol{X}_{\mathrm{G}})\right)$, where σ denotes the Sigmoid function and $\odot$ represents element-wise multiplication.

Different from naive plug-and-play stacking, the two modules form a bidirectional complementary relationship. On the computational side: GSConv compresses redundant computation, and the saved computational resources fully afford the minor extra overhead introduced by EMA; on the feature side: GSConv retains low-level structural crack features while compressing model complexity. Then EMA performs cross-spatial saliency recalibration upon $\boldsymbol{X}_{\mathrm{G}}$, amplifying feature responses of crack-related channels and suppressing interference from pavement background.

This information processing paradigm summarized as 'compress-and-preserve-first, then focus-and-enhance' constitutes the essential distinction between our coupled collaborative architecture and conventional isolated module stacking strategies.

2.5. YOLO26-GE

The overall architecture of the YOLO26-GE model, which is constructed in this work based on the aforementioned improvements, is organized following the classic three-stage 'Backbone–Neck–Head' framework of the YOLO series, as illustrated in figure 6.

In the backbone, we build upon the YOLO26 backbone. We replace the first downsampling convolutional layer with GSConv. This replacement makes shallow feature extraction lightweight. We retain the C3k2, SPPF, and C2PSA modules to preserve multi-scale feature extraction capability.

With GSConv embedded, the model reduces computational load and preserves the fine texture features of cracks. As a result, it supplies high-quality input for subsequent fusion.

In the neck, we adopt a PAN-FPN bidirectional feature pyramid structure. It combines top-down upsampling paths, bottom-up cross-layer connections, and the C3k2 module. This structure realizes bidirectional interactive fusion of multi-scale features. It balances semantic information and spatial details from different levels. Thus, it ensures robust detection for cracks of varying scales.

To improve high-level feature representation for crack targets, we embed an EMA module after the fused feature output at the lowest neck layer. The EMA module uses adaptive channel and spatial weighting. It highlights crack features and suppresses complex background interference. We then feed these enhanced features to the detect head.

This design provides YOLO26-GE with three key advantages. First, it achieves a lower parameter count and computational cost than the original YOLO26, facilitating edge-device deployment. Second, the synergy of GSConv and EMA enhances both fine-texture extraction and background suppression. Third, it improves detection accuracy without sacrificing its lightweight nature, overcoming the performance bottleneck that prevents existing models from achieving both high accuracy and high speed simultaneously.

In summary, we achieve lightweight compression through GSConv. We realize feature compensation through EMA. Through their deep synergy, we construct an improved YOLO26 model tailored for UAV road crack detection. This design provides a clear technical solution for subsequent experimental validation.

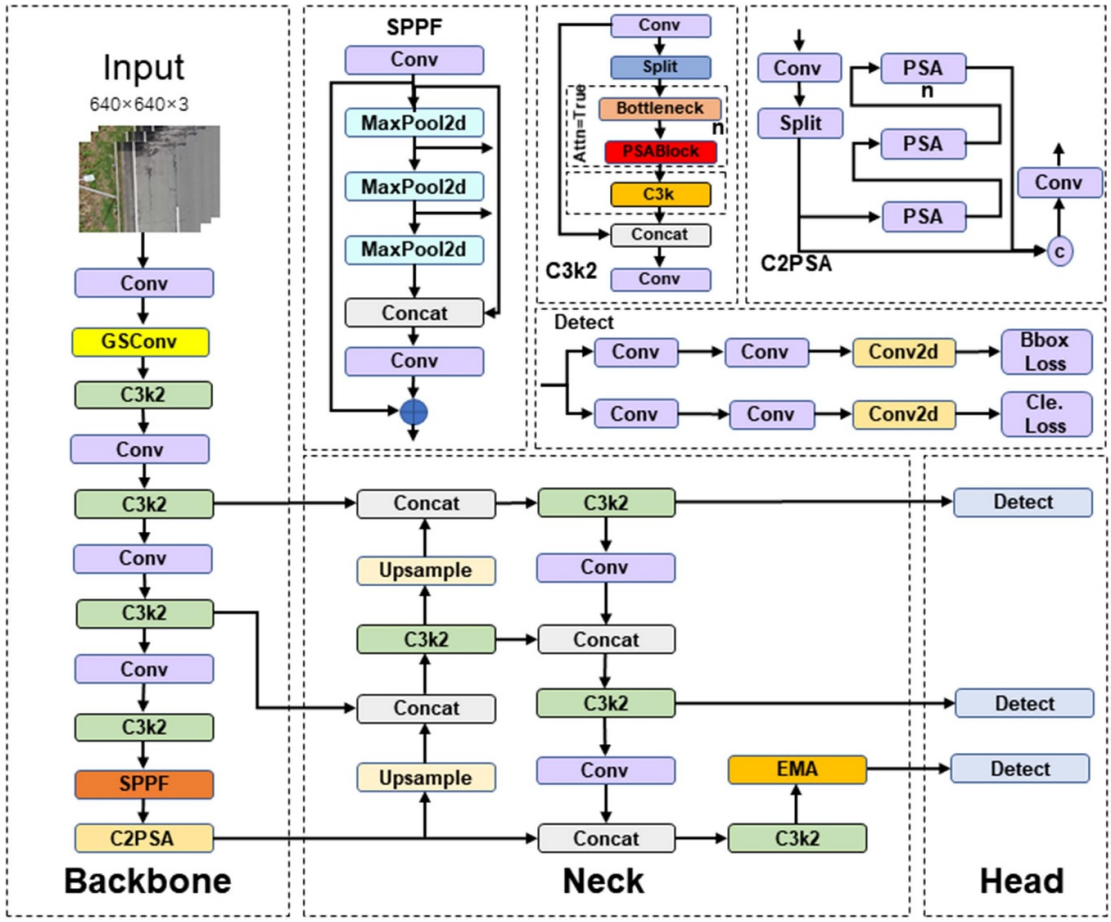


Figure 6. YOLO26-GE model network structure.

Table 1. Hardware environment configuration.

Name	Parameters
CPU	Intel Core i5-12600KF
GPU	NVIDIA GeForce RTX 4060
RAM	32GB DDR4 3200 MHz
SSD	1 TB

3. Experiments and analysis

3.1. Experimental environment

To validate the effectiveness and superiority of our proposed YOLO26-GE model for UAV-based road crack detection, we conduct a series of comparative and ablation experiments in this chapter.

We use the UAV-PDD2023 dataset as the benchmark. We clearly define the experimental environment, evaluation metrics, and experimental protocol.

Through quantitative comparisons and qualitative visualizations, we systematically validate the improvements in detection accuracy, inference speed, and model lightweightness.

At the same time, we compare our approach with existing mainstream models. These comparisons highlight the core strengths of our method. They also provide solid experimental support for its engineering deployment.

Table 1 summarizes the hardware environment for the experiments. All model training and testing were conducted on the same hardware platform to maintain consistent experimental conditions. The platform was equipped with an Intel Core i5-12600KF processor, 32 GB of RAM, and an NVIDIA GeForce RTX 4060 GPU with 8 GB of video memory. The detailed specifications are as follows:

Table 2. Software environment configuration.

Software/Library	Versions
Operating system	Windows 11
Deep learning framework	PyTorch 2.5.0
CUDA	11.8
Python	3.11
OpenCV	4.8.0
NumPy	1.26.0
Matplotlib	3.8.0

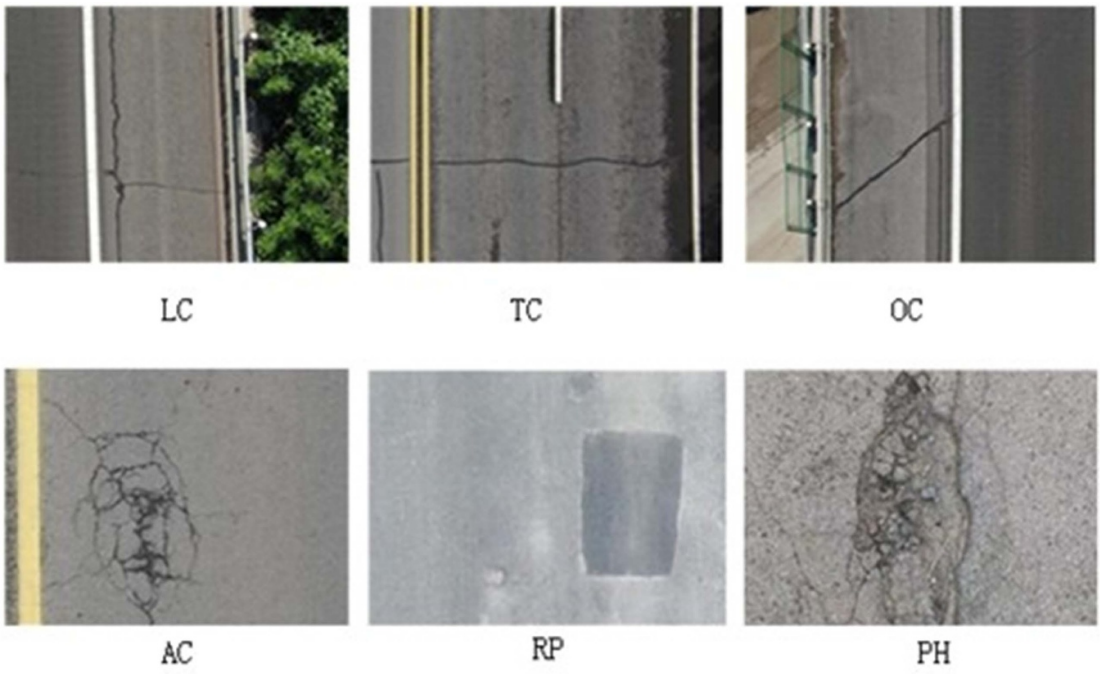


Figure 7. Dataset of road surface sample defect images.

The software environment for the experiments was built using mainstream deep learning frameworks and tools to ensure efficient model training, data processing, and result analysis. The detailed software configurations are listed in table 2.

3.2. Experimental dataset

The UAV-PDD2023 dataset [24] was employed in this study for road crack detection from UAV images. This dataset, officially designated UAV-PDD2023: a benchmark dataset for pavement distress detection Based on UAV images, is a publicly available benchmark for UAV-based pavement distress detection released by Hebei University of Technology. It comprises six typical pavement distress categories: longitudinal cracks (LC), transverse cracks (TC), oblique cracks (OC), alligator cracks (AC), repair (RP), and potholes (PH). Illustrative examples of these distress categories are presented in figure 7. Dataset of road surface sample defect images

This dataset contains 2440 UAV-acquired aerial road images. We collected these images from road networks in different regions of China. We captured the images under sunny conditions and within one hour after rainfall. They reflect real-world pavement surface states across a range of construction conditions.

All images are three-channel JPG format. Their resolution is 2594 × 1944 pixels. We captured them from a uniform altitude of 30 m.

We annotated approximately 11 158 crack instances in the PASCAL VOC format. The annotations include crack bounding box coordinates and types. The annotation accuracy exceeds 95%. This ensures reliable model training and evaluation.

Table 3. Hyperparameter setting.

Hyperparameter	Value
lr0	0.001
lrf	0.01
weight_decay	0.0005
batch	16
epochs	700

To ensure experimental rigor and reproducibility, we randomly partitioned the UAV-PDD2023 dataset. We divided it into training, validation, and test sets at a ratio of 7:2:1. The training set has 1708 images, the validation set has 488 images, and the test set has 244 images.

Crack objects in UAV aerial images are typically elongated. They exhibit low contrast. They are also prone to background interference.

To address these issues, we designed a tailored preprocessing and data augmentation strategy. This strategy enhances model generalization and detection accuracy. It also preserves crack structural information.

(1) Basic preprocessing

Size normalization: all input images were resized to 640 × 640 pixels while maintaining their original aspect ratio, with padding applied where necessary.

Pixel normalization: image pixel values were normalized to [0, 1] to accelerate convergence and improve training stability.

(2) Targeted data augmentation

Based on the characteristics of pavement cracks, destructive augmentations were suppressed, and structural information was preserved. The specific augmentation parameters and strategies were as follows:

Mosaic augmentation (probability 0.5):

A four-image Mosaic strategy was applied with a moderate probability to avoid excessive cropping or blurring of fine, low-texture cracks, while improving generalization to complex road scenes.

Mixup augmentation disabled:

Pavement cracks are sparse small objects; mixup can cause label ambiguity and boundary blurring, and was therefore disabled (set to probability 0.0).

Mosaic disabled at the end of training:

Mosaic augmentation was turned off during the last 20 epochs (epochs 681–700) to eliminate stitching artifacts, enabling the model to learn authentic pavement structural features and achieve stable convergence.

Lightweight HSV color augmentation: a low-intensity HSV augmentation was employed with hue shift hsv_h = 0.01 (extremely weak), saturation hsv_s = 0.3, and value hsv_v = 0.3. This moderate, low-intensity strategy preserved crack structure while improving robustness to varying illumination conditions.

(3) Training hyperparameters and optimization

The AdamW optimizer was employed for model training to suit the convergence behavior of small-object detection. The hyperparameters, including the learning rate and regularization strategy, are listed in table 3.

3.3. Experimental evaluation indicators

To comprehensively evaluate model performance on crack detection, the following metrics are adopted.

Precision (P) is defined as the proportion of true crack instances among all instances predicted as cracks, thereby measuring the model's capability to suppress FP, and is given by:

$$P = \frac{\mathrm{TP}}{\mathrm{TP+FP}} \quad . \tag{1}$$

Recall (R) is defined as the proportion of ground-truth crack instances that are correctly detected, thereby measuring the capability to suppress FN, and is given by:

$$R = \frac{\mathrm{TP}}{\mathrm{TP+FN}} \quad . \tag{2}$$

In the definition, TP (true positive) denotes the number of correctly detected crack instances, FP denotes the number of non-crack instances incorrectly predicted as cracks, and FN denotes the number of crack instances left undetected. Higher recall reflects improved identification of fine and low-contrast cracks.

$F1$-score ($F1$) is defined as the harmonic mean of precision and recall, thereby providing a balanced measure of detection accuracy, and is given by:

$$F1 = \frac{2PR}{P+R} \quad . \tag{3}$$

A more optimal balance between precision and recall is achieved as the $F1$-score approaches 1, thereby better satisfying the requirement for simultaneous suppression of FP and FN in UAV-based crack inspection.

We define mAP@0.5 as the mean average precision over all crack categories at an intersection over union (IoU) threshold of 0.5. This metric serves as a core indicator for evaluating overall model performance in object detection. It comprehensively reflects the model's ability to recognize cracks of various types and scales. A higher mAP@0.5 value means better overall detection performance.

We define mAP@0.5:0.95 as the average mAP computed over IoU thresholds from 0.5 to 0.95 in steps of 0.05. Unlike mAP@0.5, which uses a single IoU threshold, this metric more fully characterizes the model's localization and recognition accuracy across different overlap requirements

$$AP = \int_0^1 P(R)\,\mathrm{d}R \tag{4}$$

$$\mathrm{mAP} = \frac{\sum_{i=1}^{K} AP_i}{K} \quad . \tag{5}$$

Inference speed, measured in frames per second (FPS), denotes the number of images the model processes per second. It directly reflects the model's real-time inference capability. This metric is a key indicator for meeting the engineering demands of long-distance and rapid UAV inspection. We compute it as:

$$\mathrm{FPS} = \frac{1}{t} \quad . \tag{6}$$

We define t*t* as the average processing time per image (in seconds). Higher FPS means stronger real-time inference capability.

Lightweight metrics evaluate the model's deployment potential on resource-constrained UAV edge devices. We define the number of parameters (Parameters) as the total count of trainable parameters. Fewer parameters reduce storage requirements and facilitate edge deployment. We define computational cost (GFLOPs) as the number of floating-point operations required for a single forward pass. We express it in gigaFLOPs (10^9 FLOPs). Lower GFLOPs demand less computational power. This makes the model more suitable for efficient inference on resource-constrained UAV platforms.

3.4. Experimental plan design

This study comprehensively validates the GSConv–EMA synergy through six experiment types: ablation studies, comparative experiments, cross-dataset generalization, qualitative visualization, feature map visualization, and failure case analysis. These experiments assess the method from six perspectives—module mechanisms, performance comparison, scene robustness, visual recognition, feature learning behavior, and model limitations.

Ablation experiments. We adopt YOLO26n as the baseline and build model variants to isolate GSConv and EMA contributions and examine their coupling effect. This step separates performance gains brought by lightweight compression and feature enhancement, quantifies synergistic benefits, and elaborates the shallow-lightweight, deep-compensation mechanism.

Comparative experiment. We compare YOLO26-GE with baseline YOLO26 and mainstream lightweight detectors—YOLOv5n, YOLOv8n, and YOLOv11n—under identical datasets, hardware, and hyperparameters. We evaluate accuracy, speed, and model size via parameters, GFLOPs, $F1$, mAP@0.5, and mAP@0.5:0.95. Results verify the model's excellent comprehensive trade-off.

Cross-dataset generalization experiment. To avoid overfitting on a single aerial dataset, we establish cross-dataset validation to evaluate generalization across diverse road scenes. With fixed architectures and hyperparameters, YOLO26-GE and baseline YOLO26 are trained from scratch and tested on three

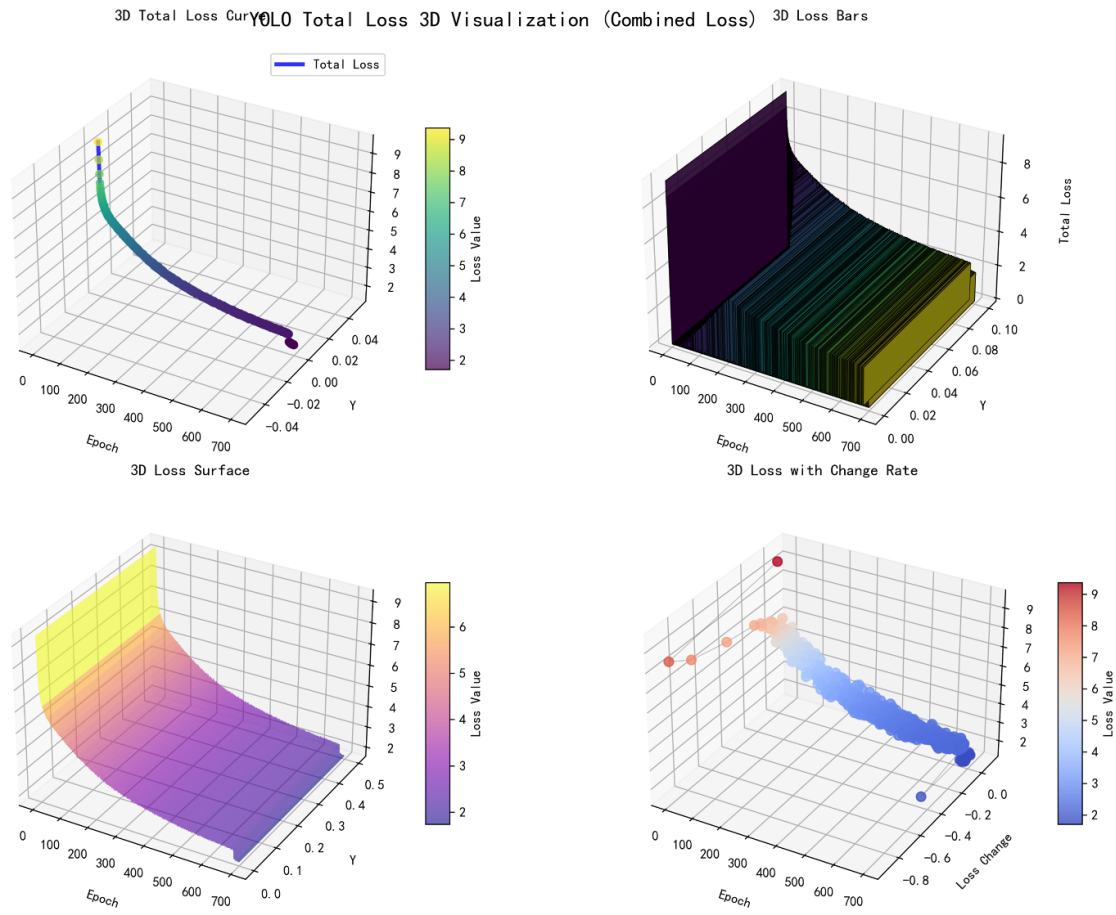


Figure 8. Total training loss 3D multi-dimensional visualization graph.

datasets with distinct perspectives and crack forms: UAV-PDD2023 (aerial), RDD2022 (urban road damage), and HighRPD (ground high-resolution). Metric fluctuations across domains confirm the model's universality.

Qualitative visualization. We select typical road damages including LC, TC, AC, and potholes to compare detection outputs and Grad-CAM heatmaps. This comparison evaluates detection integrity, bounding box precision and activation focus, demonstrating enhanced detection of low-contrast faint cracks and supporting quantitative results.

Feature map visualization. We extract shallow and high-level feature maps from four ablation models for comparison. Visual results verify bidirectional synergy: GSConv eliminates redundant features, while EMA enhances crack responses and suppresses background noise, offering intuitive evidence for module cooperation.

Failure case analysis. We analyze failure samples to explore remaining limitations, focusing on challenging scenarios: damp dark pavement after rainfall and ultra-thin cracks with tiny pixel occupancy, to diagnose defective feature extraction. This analysis identifies model constraints and indicates future optimization directions.

3.5. Experimental results and analysis

3.5.1. Analysis of the model training process

Figure 8. Shows the three-dimensional visualization of the total loss during YOLO26-GE model training. It includes four subplots: loss curve, loss surface, histogram of loss values, and loss change rate analysis. The loss curve in the upper left shows that the total loss decreases steadily as training progresses. It converges from an initially high value to approximately 2.0. It shows no notable oscillations or divergence.

The loss surface in the lower left further confirms the smooth descent of the loss. It exhibits minimal fluctuations. This indicates stable training.

The histogram of loss values (upper right) and the loss change rate analysis (lower right) demonstrate that the loss decreases rapidly during the early stage of training. It then levels off gradually in

Table 4. Comparison of ablation study results.

Model	GFLOPs	*P*	*R*	*F*1	mAP@50	mAP@0.5:0.95	FPS
Option A	5.2	0.754	0.618	0.679	0.677	0.369	769
Option B	5.1	0.732	0.687	0.709	0.729	0.433	769
Option C	5.1	0.813	0.639	0.716	0.709	0.394	769
Option D	4.7	0.739	0.673	0.704	0.722	0.438	417
Option E	5.2	0.722	0.651	0.685	0.695	0.398	714
Option F	5.2	0.746	0.638	0.688	0.681	0.400	769
Option G	4.5	0.772	0.650	0.705	0.696	0.410	667
Option H	5.1	0.683	0.607	0.642	0.657	0.376	714
Option I	5.1	0.762	0.654	0.704	0.690	0.389	714
Option J	5.2	0.759	0.592	0.665	0.657	0.360	667
Option K	5.1	0.730	0.643	0.683	0.700	0.404	714
Option L	**5.1**	**0.792**	**0.688**	**0.736**	**0.728**	**0.413**	**714**

the middle and later stages. Finally, it converges to a stable value. There is no evidence of overfitting or underfitting.

These results indicate that our proposed model achieves stable training and favorable convergence. This establishes a reliable foundation for the subsequent performance improvement.

3.5.2. Results and analysis of ablation experiments

To clarify the specific contribution of each improvement measure to the overall performance of the model and systematically explore the optimal embedding positions of the GSConv lightweight convolution and EMA multi-scale attention mechanisms, we designed a series of candidate network variants based on the original YOLO26n baseline. All options are numbered as Option A to Option K for intuitive comparison. Option A represents an unmodified baseline model with no optimization at all. Options B, C and D are used to explore the performance differences brought about by GSConv under different single-layer, last-layer and full backbone replacement strategies without attention modules. Option F only introduces EMA attention at the P5 layer to analyze the independent gain of the attention block. Scheme E and G–K simultaneously integrate GSConv and EMA, and further conduct comparative ablation from two dimensions: the replacement position of GSConv in the backbone and the embedding position of EMA in the neck (P3, P4, P5, multi-layer joint embedding, backbone tail embedding). After a comprehensive comparison of the FPS, precision, recall, *F*1-score and mAP@0.5 metrics of all candidate architectures, option K achieved the best comprehensive detection performance and was identified as the final YOLO26-GE model proposed in this paper. The results are presented in table 4. The bold values in table 4 represent the best performance of each metric among all compared models.

To explore the independent effectiveness of the GSConv lightweight convolution and EMA attention mechanism, as well as determine their optimal deployment locations, multiple network variants are constructed on the basis of the YOLO26n baseline (Option A). Quantitative results of all schemes are summarized in table 4. Option B–D are designed to explore different GSConv replacement strategies without attention modules; Option F only introduces the EMA module to evaluate the independent performance of the attention mechanism. Option E, G–L combine GSConv and EMA simultaneously, covering various GSConv replacement positions in the backbone and different embedding locations of EMA in the neck.

The ablation results demonstrate that either GSConv or EMA can improve the detection performance of the baseline model, and the collaborative combination of GSConv and EMA yields prominent mutual complementation, achieving the optimal overall performance.

Independent effect of GSConv: compared with Option A (baseline), replacing SC with GSConv brings varying performance gains. For Option B, where only the second backbone convolution is substituted by GSConv, the inference speed remains 769 FPS. In terms of detection metrics, the recall (R) rises substantially from 0.618 to 0.687 (+11.2%), mAP@0.5 increases from 0.677 to 0.729 (+7.7%), and the $F1$-score improves from 0.679 to 0.709 (+4.4%). Nevertheless, precision (P) drops slightly from 0.754 to 0.732 (−2.9%). These results reveal that GSConv eliminates redundant computation of SC while preserving fine texture and edge features of pavement cracks, significantly reducing missed detections of challenging targets such as thin and OC. The slight precision degradation originates from the limited inter-channel information exchange inherent in group convolution, which causes partial background regions to be falsely identified as cracks. This drawback can be effectively compensated by introducing the EMA attention mechanism.

Independent effect of EMA attention mechanism: As shown in Option F, embedding the EMA module alone improves recall from 0.618 to 0.638 (+3.2%), and mAP@0.5 increases from 0.677 to 0.681 (+0.6%). The improvement verifies that EMA strengthens the feature focusing capability of the model and mitigates feature loss caused by lightweight operations. By adaptively recalibrating channel-wise feature responses, EMA dynamically enhances crack-related feature channels and suppresses background noise. However, the attention module introduces extra computational overhead. Simply stacking EMA cannot achieve satisfactory precision improvement, and this limitation can be resolved via combination with GSConv.

Synergistic effect of GSConv and EMA: among all combined schemes, Option L (the proposed YOLO26-GE) achieves the best trade-off between detection accuracy and inference speed. It obtains the highest precision of 0.792 (+5.0% compared with the baseline), which fully offsets the precision decline observed in Option B and surpasses all single-module variants. Its recall reaches 0.688 (+11.3%), comparable to Option B. Meanwhile, the *F*1-score of 0.736 (+8.4%) and mAP@0.5 of 0.728 (+7.5%) outperform all other architectures.

In terms of computational cost, Option L maintains 5.1 GFLOPs, which is close to the lightweight GSConv-only scheme. The inference speed is 714 FPS. Although it is lower than the 769 FPS of Option B, it exceeds most combined schemes and meets the real-time detection requirements for UAV edge deployment. The FPS reduction compared with Option B comes from the forward computation overhead introduced by EMA. Benefiting from GSConv replacing the SC at the second layer of backbone, the computational burden is reduced, which offsets the extra cost of the attention module.

Comparing other combined configurations (Option E, G–K), embedding EMA on P3, P4 or multiple feature layers cannot deliver better comprehensive performance. Deploying EMA on shallow feature layers (Option H) leads to obvious performance degradation. Multi-layer joint embedding of EMA (Option J) increases computation and causes feature redundancy. Only embedding EMA on the high-semantic P5 layer can maximize the complementary advantages of GSConv and EMA.

In summary, GSConv and EMA form mutually reinforced optimization. GSConv reduces computational complexity and alleviates missed detection, while EMA suppresses FP and compensates for channel information loss induced by group convolution. When deployed with the optimal layout in Option L, the model can detect more crack targets accurately and restrain background interference simultaneously, realizing comprehensive performance promotion.

Each option corresponds to an independent network improvement scheme based on the YOLO26n baseline (Option A). The detailed modification of each architecture is elaborated as follows:

Option A: Original YOLO26n baseline model without any module replacement or attention embedding.

Option B: Only the second SC layer in the Backbone is replaced with the GSConv lightweight module, and no attention mechanism is introduced.

Option C: Only the last SC layer of the backbone is substituted by GSConv, without embedding any attention block.

Option D: All SC layers throughout the entire backbone are replaced with GSConv, and no attention module is added.

Option E: The first convolution layer of the backbone is replaced with GSConv, and the EMA attention mechanism is embedded on the P5 feature layer of the neck branch simultaneously.

Option F: The EMA multi-scale attention module is only deployed on the neck P5 feature layer, while all original convolution layers of backbone remain unchanged.

Option G: All convolution layers of the backbone are fully replaced by GSConv, and the EMA module is additionally inserted on the neck P5 feature layer.

Option H: The second convolution layer of the backbone is replaced with GSConv, and the EMA attention block is embedded on the neck P3 shallow feature layer.

Option I: The second convolution layer of the backbone is replaced with GSConv, and the EMA attention block is embedded on the neck P4 middle feature layer.

Option J: The second convolution layer of the backbone is replaced with GSConv, and EMA attention modules are synchronously embedded on three feature layers P3, P4 and P5 of the Neck.

Table 5. Performance comparison of different models.

Model	Parameters	GFLOPs	*F*1	mAP@50	mAP@50:95
YOLOv3-tiny	12.5 M	18.9	0.557 ± 0.018	0.519 ± 0.020	0.233 ± 0.013
YOLOv5n	2.51 M	7.1	0.567 ± 0.015	0.519 ± 0.020	0.308 ± 0.015
YOLOv6n	4.24 M	11.8	0.648 ± 0.015	0.621 ± 0.016	0.330 ± 0.014
YOLOv8n	3.00 M	8.1	0.659 ± 0.014	0.628 ± 0.015	0.322 ± 0.013
YOLOv9-tiny	2.0 M	7.9	0.666 ± 0.013	0.649 ± 0.014	0.351 ± 0.012
YOLOv10n	2.7 M	8.4	0.677 ± 0.021	0.672 ± 0.015	0.371 ± 0.015
YOLOv11n	2.59 M	6.4	0.699 ± 0.012	0.692 ± 0.013	0.382 ± 0.011
YOLO26	2.51 M	5.2	0.698 ± 0.019	0.705 ± 0.025	0.389 ± 0.019
YOLO26-GE	**2.51 M**	**5.1**	**0.724 ± 0.015**	**0.722 ± 0.008**	**0.415 ± 0.005**

Option K: Replace the second convolutional layer of the backbone with GSConv, and move the EMA module to the backbone.

Option L (Ours, final proposed YOLO26-GE model): The second convolution layer of the backbone is replaced with GSConv, and a single EMA attention module is embedded on the neck P5 high-level semantic feature layer.

3.5.3. Comparative experimental results and analysis

To quantitatively evaluate the comprehensive performance gap between our improved model and existing mainstream lightweight detectors, we conducted horizontal comparative experiments. We selected nine mainstream lightweight models as benchmarks: YOLOv3-tiny, YOLOv5n, YOLOv6n, YOLOv8n, YOLOv9-tiny, YOLOv10n, YOLOv11n, the original YOLO26, and YOLO26-GSConv (which replaces SC with GSConv). To ensure fairness, all models were trained and tested using the identical hardware environment, UAV-PDD2023 dataset, and hyperparameter settings. Results are shown in table 5. The bold values in table 5 represent the best performance of each metric among all compared models. To verify statistical reliability, we repeated the training and testing using three different random seeds (0, 42, 123), and report the mean and standard deviation for all metrics. Among all compared algorithms, our YOLO26-GE achieves the optimal balance between detection precision and lightweight design. It attains the highest scores across all three core metrics: *F*1 (0.724 ± 0.015), mAP@0.5 (0.722 ± 0.008), and mAP@0.5:0.95 (0.415 ± 0.005). Furthermore, it maintains a compact size of only 2.5 M parameters and 5.1 GFLOPs, highlighting its clear advantage in efficiency. Compared with the baseline YOLO26, our model reduces inference overhead while delivering substantial accuracy gains: *F*1 improves from 0.698 ± 0.019 to 0.724 ± 0.015 (a 3.72% increase), mAP@0.5 improves from 0.705 ± 0.025 to 0.722 ± 0.008 (+2.41%), and mAP@0.5:0.95 improves from 0.389 ± 0.019 to 0.415 ± 0.005 (+6.68%). Moreover, the consistently low standard deviations across all metrics confirm that these improvements are statistically stable and not occasional fluctuations from a single trial. Other lightweight models suffer from inherent limitations: YOLOv3-tiny lacks sufficient accuracy due to its oversimplified structure; YOLOv6n imposes excessive computational costs, precluding onboard real-time inference; and YOLOv9-tiny, YOLOv10n, and YOLOv11n provide only moderate performance, failing to match YOLO26-GE in both accuracy and efficiency. In general, these existing lightweight models have weak feature representation capabilities, making it difficult to extract low-contrast, fine-grained crack textures from aerial UAV images. In conclusion, YOLO26-GE achieves an ideal balance among parameter count, computational cost, and detection accuracy, making it particularly suitable for routine UAV-based road crack inspection in practical engineering scenarios.

3.5.4. Generalization experiment results and analysis

To determine whether the performance gains of YOLO26-GE are dataset-dependent and to rule out experimental bias, we trained and evaluated our model on three diverse public pavement crack datasets: UAV-PDD2023, HighRPD, and RDD2022. We trained all models from scratch independently on each dataset. Table 6 summarizes the unified evaluation metrics, covering precision, recall, *F*1-score, mAP@0.5, and mAP@0.5:0.95.

The three datasets exhibit notable distribution differences. UAV-PDD2023 contains aerial pavement images. HighRPD provides ground-level distress data from different viewpoints. RDD2022 poses greater challenges with complex backgrounds and varied crack patterns.

On UAV-PDD2023, YOLO26-GE consistently outperforms the baseline YOLO26: precision rises from 0.754 to 0.792, recall from 0.618 to 0.688, *F*1-score from 0.679 to 0.736, and mAP@0.5 from 0.677 to

Table 6. Generalization experiment.

Datasets	Model	*P*	*R*	*F*1	mAP@50	mAP@50:95
UAV-PDD2023	YOLO26	0.754	0.618	0.679	0.677	0.369
	YOLO26-GE	0.792	0.688	0.736	0.728	0.413
RDD2022	YOLO26	0.608	0.51	0.554	0.524	0.258
	YOLO26-GE	0.620	0.524	0.568	0.549	0.272
HighRPD	YOLO26	0.633	0.588	0.610	0.588	0.299
	YOLO26-GE	0.655	0.614	0.634	0.614	0.312

0.728. Parameter count also drops slightly from 5.2 M to 5.1 M, confirming that GSConv and EMA attention jointly strengthen feature representation for faint, narrow cracks while reducing model size.

On HighRPD, which differs from UAV-PDD2023 in sample distribution and imaging style, YOLO26-GE continues to outperform the baseline, raising mAP@0.5 from 0.588 to 0.614 alongside improvements in precision, recall, and *F*1-score. These consistent gains across datasets confirm the general effectiveness of our proposed modules.

On the more challenging RDD2022, YOLO26-GE achieves an mAP@0.5 of 0.549, surpassing the baseline (0.524), and also delivers higher precision, recall, and *F*1-score. Although RDD2022 presents a lower overall detection baseline, our improvements still yield stable and consistent gains.

From these multi-dataset experiments, we draw the following key conclusions:

First under independent full-training setups on all three diverse datasets, YOLO26-GE consistently surpasses the YOLO26 baseline. This confirms that the gains from the GSConv-EMA combination are not accidental or dataset-restricted, and that our architecture exhibits strong generalizability.

Second, the performance variations of YOLO26 across datasets stem from inherent discrepancies in sample composition, crack morphology, background clutter, and annotation standards, not from intrinsic deficiencies of the baseline model. Third, GSConv enables lightweight feature encoding with only a slight parameter adjustment, while EMA selectively amplifies feature responses for crack targets. Their combined optimization remains effective across diverse data distributions, demonstrating that our scheme generalizes well to various road crack detection scenarios. We note that this study validates the GSConv-EMA framework only under separate full-training conditions on different datasets. Zero-shot cross-domain inference under significant domain shifts will be addressed in our future work.

3.5.5. Analysis of visualization results

For qualitative validation of the proposed model's detection performance, test set samples were selected and its predictions were compared with those of the original YOLO26. The visual comparisons are presented in figure 9. The visual comparisons demonstrated that the proposed model markedly outperformed the original YOLO26 in detection performance. As shown in a comparison of figure 9(a) and figure 9(b), YOLO26 GE detected more distress instances than the original model, and no missed detections were observed. First, YOLO26 GE achieved consistently higher prediction confidence across various distress types (including alligator, transverse, oblique, and LC, as well as potholes), yielding greater certainty in identifying critical distress targets.

Second, the baseline YOLO26 exhibited notable missed detections and an insufficient capability to recognize crack features in the presence of weak textures and complex backgrounds, whereas YOLO26 GE detected distress instances more comprehensively, with bounding boxes better fitted to the actual distress contours. Furthermore, YOLO26 GE effectively suppressed FP induced by background noise and crack like textures, demonstrating superior feature discrimination and robustness against interference. These qualitative findings are consistent with the improvements observed in the quantitative metrics (precision, recall, and mAP@0.5) reported above, thereby confirming the efficacy of the proposed modules in enhancing distress detection performance. To validate the effectiveness and interpretability of the model's decisions, the Grad CAM method was employed to generate feature activation visualizations for the YOLO26 GE model. The results are presented in figure 10.

'Original' denotes the input pavement distress image, 'Heatmap' denotes the class activation heatmap, and 'Overlay' denotes the fused heatmap and original image. Warmer colors (red/yellow) indicate regions of higher model attention. As shown in figure 10, the high activation regions were precisely localized on pavement distresses—including AC, TC, and potholes—without interference from irrelevant background elements (e.g. manhole covers and road textures). These observations confirm that the YOLO26 GE model effectively attends to key distress features rather than depending on background noise or spurious features for prediction, thus validating the soundness of its decision logic and the effectiveness of its

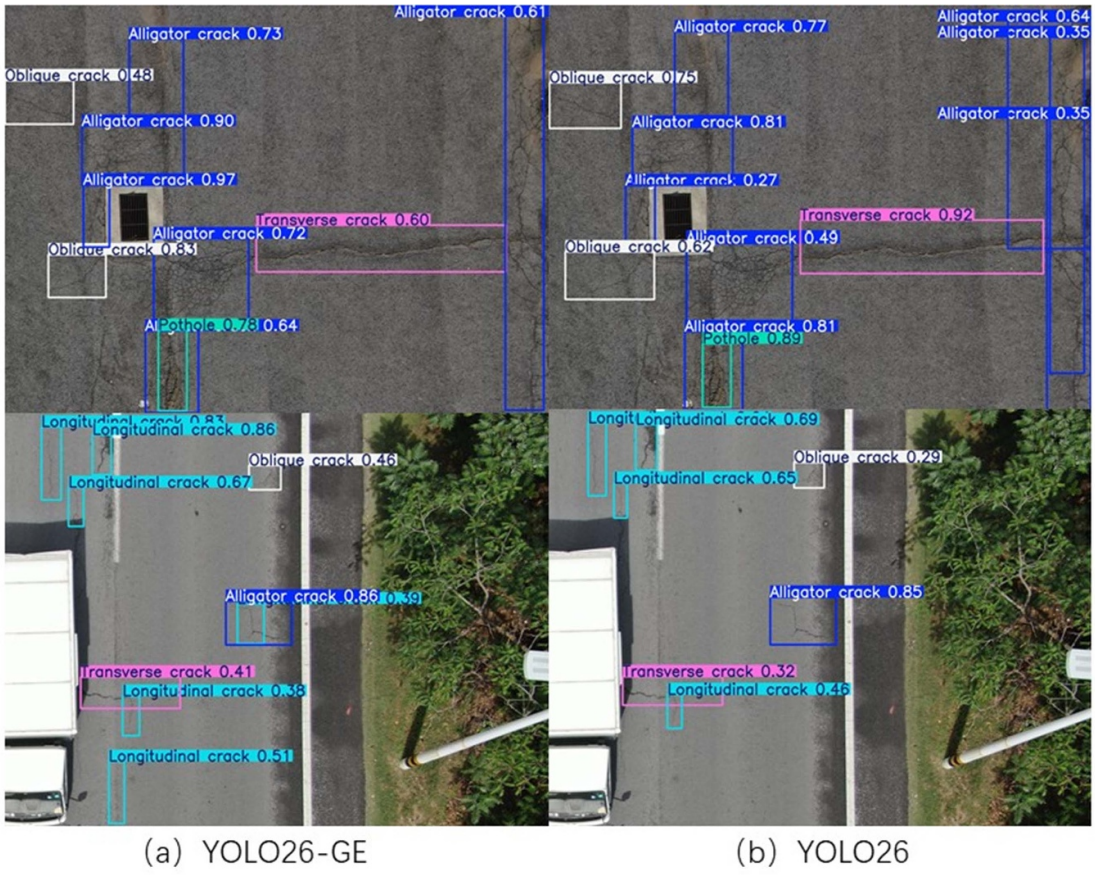


Figure 9. Comparison of visualization results.

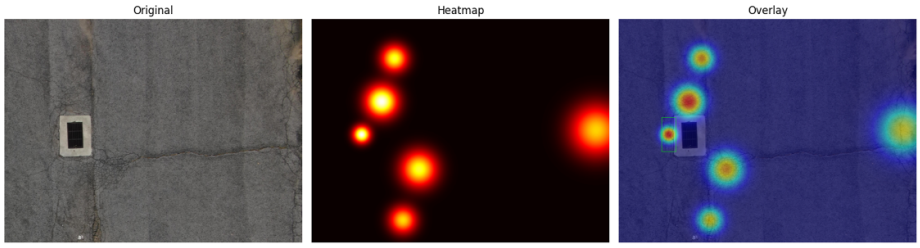


Figure 10. Heatmap of feature activation regions of the YOLO26-GE model.

feature learning. This also provides interpretability support for the quantitative improvements reported earlier.

3.5.6. Failure-case analysis

To objectively evaluate the practical limitations of the YOLO26-GE model, this section analyzes typical missed-detection samples from the UAV-PDD2023 test set, as shown in figure 11. Green boxes denote ground-truth crack annotations, while blue boxes denote model prediction outputs.

As depicted in figures 11(a) and (b), these samples are captured after rainfall. The wet pavement presents a darker overall tone, which greatly reduces the grayscale contrast between cracks and road background and obscures crack textures. Combined with heavy tree shadows, valid crack features are further suppressed and missed detections are triggered. Figures 11(c) and (d) illustrate detection results on dry-surface roads. Although cracks obtain higher contrast against dry pavement, local missed detections and fragmented prediction boxes still frequently occur when confronting numerous complex, discontinuous tiny cracks.

Although the joint design of GSConv and EMA achieves a favorable trade-off between inference cost and detection accuracy, the model still suffers from inherent limitations. Its feature representation

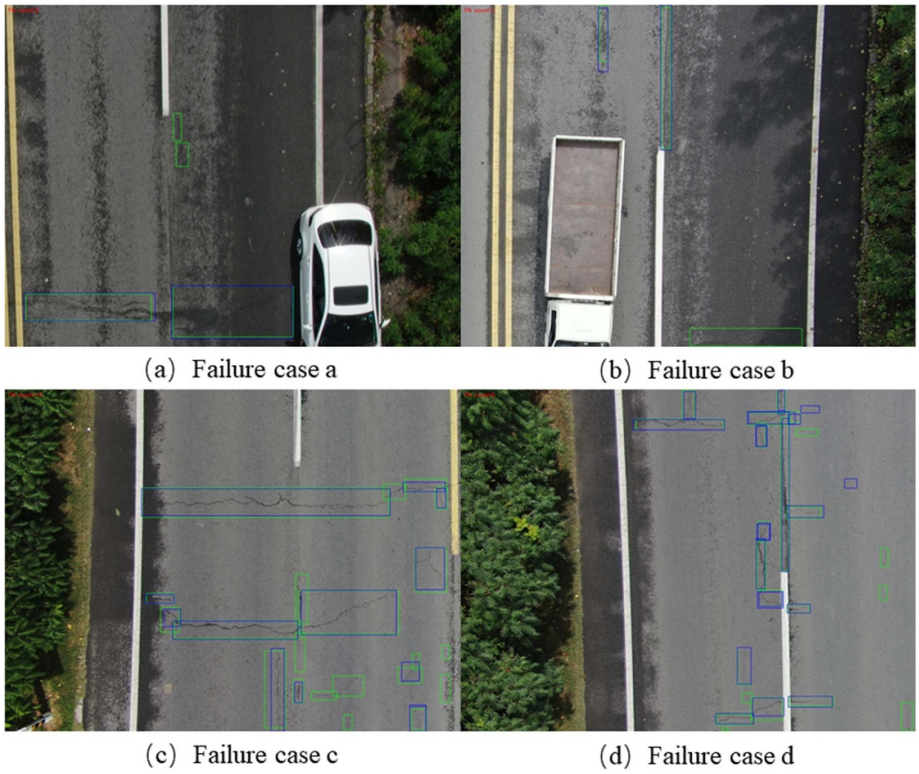


Figure 11. YOLO26-GE model's typical failure cases in the UAV-PDD2023 test set. The green boxes represent the actual crack annotations, and the blue boxes represent the model's detection outputs. Four sub-figures show two types of typical missed detection scenarios: low-texture blurred cracks and tiny fine cracks.

degrades significantly under conditions such as post-rain wet pavement, heavy shadow interference, and ultra-fine cracks with extremely low pixel occupation. For UAV aerial-view tasks, long-distance shooting further shrinks the effective pixels of tiny cracks and aggravates the above defects.

3.5.7. Feature visualization and module synergy mechanism analysis

To intuitively analyze the feature-learning behavior of each module and verify the synergistic effect between GSConv and EMA, we conduct feature visualization on representative feature maps of four models. We select the shallow C3k2 layer 2 and the high-level pre-detect feature layer for comparison. The visualization results are shown in figure 12.

Figures 12(a) and (b) are feature maps of the original YOLO26 baseline. The shallow layer can roughly capture crack contour information, but it responds strongly to background textures. The high-level features produce scattered activation regions. Many background areas are falsely activated, and the response to real crack targets is dispersed. This phenomenon explains the high FP rate of the baseline model in UAV-captured road crack detection tasks.

Figures 12(c) and (d) show the outputs of the only-GSConv model. After lightweight convolution replaces the original convolution, shallow feature responses become sparser. GSConv reduces redundant feature computation and compresses model parameters. However, this lightweight operation loses part of fine-grained crack detail information. The high-level feature still generates interference activation on complex backgrounds, and the target-related response is not sufficiently concentrated.

Figures 6(e) and (f) present the visualization results of the only-EMA model. The EMA attention module suppresses partial background noise and enhances target-related feature responses. Nevertheless, the original heavy-weight convolution brings massive redundant feature representations. The attention module cannot fully filter useless information, so some background interference still remains in feature maps.

Figures 12(g) and (h) correspond to the proposed YOLO26-GE model, which combines GSConv and EMA. In shallow layers, GSConv implements efficient lightweight feature extraction and eliminates redundant feature components. On this basis, EMA further strengthens feature responses toward crack regions and suppresses background noise. In high-level pre-detect features, activation energy is highly concentrated on real crack targets. False activation from road surface textures is obviously suppressed.

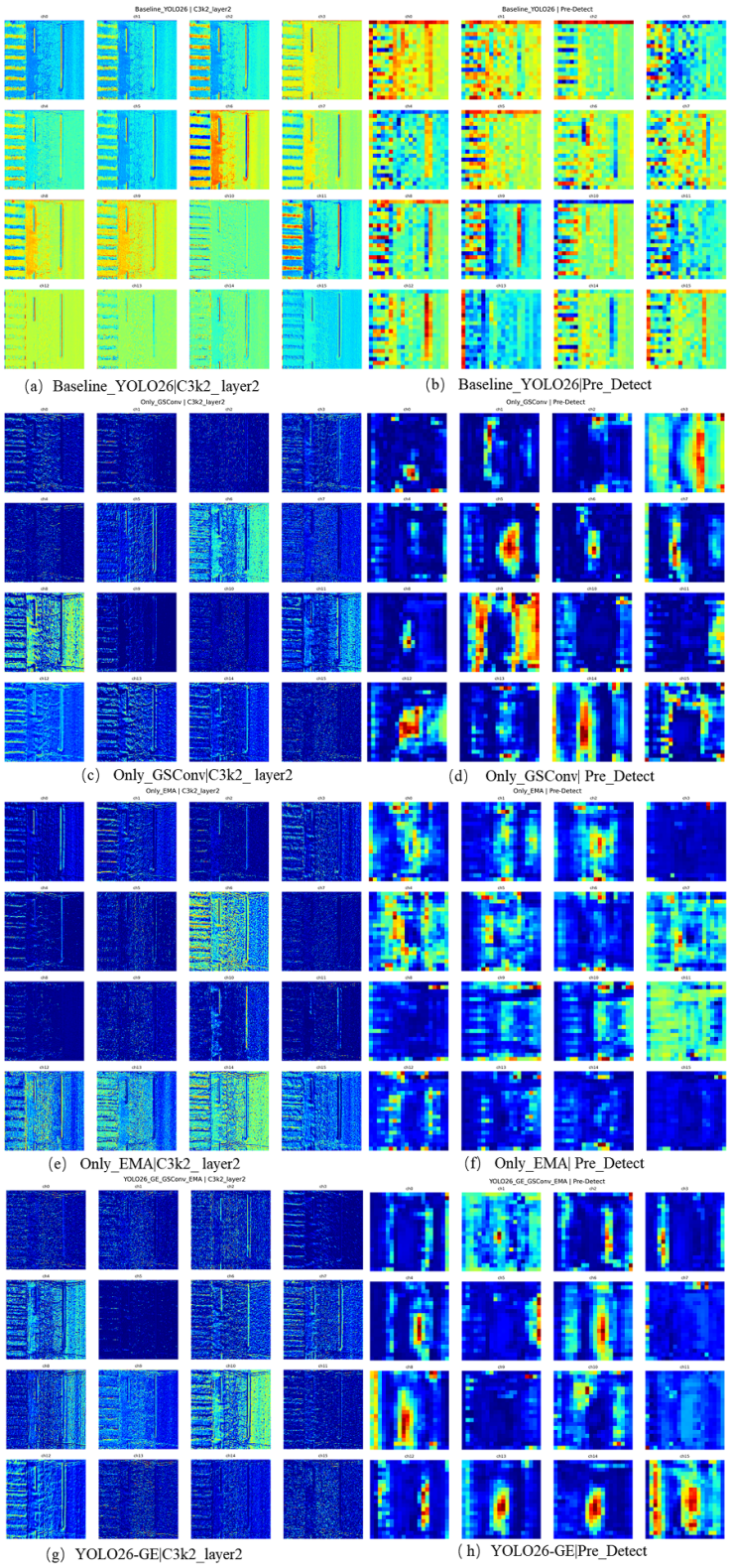


Figure 12. Feature visualization comparison of different ablation models. (a) Baseline-YOLO26, shallow C3k2 layer 2; (b) baseline-YOLO26, pre-detect layer; (c) only-GSConv, shallow C3k2 layer 2; (d) only-GSConv, pre-detect layer; (e) only-EMA, shallow C3k2 layer 2; (f) only-EMA, pre-detect layer; (g) YOLO26-GE, shallow C3k2 layer 2; (h) YOLO26-GE, pre-detect layer. All feature maps adopt the jet colormap.

The above observation proves the bidirectional synergy between GSConv and EMA. GSConv removes redundant feature representations and provides high-quality compressed features for the attention module. EMA focuses effective feature weights on crack targets and compensates for fine-grained detail loss introduced by lightweight convolution.

The above observation proves the bidirectional synergy between GSConv and EMA. GSConv removes redundant feature representations and provides high-quality compressed features for the attention module. EMA focuses effective feature weights on crack targets and compensates for fine-grained detail loss introduced by lightweight convolution. The cooperation of two modules reduces missed detection and false detection simultaneously, and finally improves detection accuracy under a low-parameter budget.

4. Conclusion

To address the core trade-off between lightweight inference and detection accuracy in UAV-based road crack detection, we take the lightweight NMS-free baseline YOLO26 as our foundation and propose YOLO26-GE, a collaborative detection model that integrates GSConv lightweight convolution and EMA multi-scale attention.

We conducted multi-dimensional experiments on the public UAV-PDD2023 dataset. In horizontal comparisons, we benchmarked nine mainstream lightweight YOLO models under identical hardware, dataset, and hyperparameter settings, with multiple repeated trials. Results show that YOLO26-GE has only 2.51 M parameters and 5.1 GFLOPs, and achieves an inference speed of 714 FPS. Compared with the original YOLO26, it improves $F1$ by 3.72%, mAP@0.5 by 2.41%, and mAP@0.5:0.95 by 6.68%. It outperforms other lightweight detectors including YOLOv3-tiny, YOLOv5n, YOLOv8n, and YOLOv11n across model size, speed, and accuracy simultaneously.

In ablation studies, we systematically investigated the individual and combined effects of GSConv placement and EMA insertion depth. Using GSConv alone reduces computational cost and misses fewer fine cracks, but causes a slight accuracy drop due to channel-wise information loss from grouped convolution. Using EMA alone strengthens weak crack responses and suppresses background noise, yet introduces extra inference overhead. However, deploying GSConv in the shallow backbone and EMA at the high-level P5 neck layer forms a complementary pair: it reduces both missed detections and FP without significantly increasing computation. This validates the synergistic benefit of our 'lightweight compression in shallow layers, feature compensation in high layers' design.

Cross-dataset generalization experiments were conducted on three distinct pavement distress datasets—UAV-PDD2023, RDD2022, and HighRPD—which differ significantly in imaging viewpoint and scene complexity. YOLO26-GE consistently surpasses the baseline YOLO26 on all three datasets, with concurrent improvements across all accuracy metrics. This demonstrates that the joint GSConv-EMA strategy is not limited to aerial images, and generalizes well to ground-level and complex urban road crack scenarios.

Combining all quantitative metrics with qualitative analyses from Grad-CAM heatmaps, detection visualizations, and typical failure cases, we conclude that YOLO26-GE, through deep coupling of lightweight convolution and efficient attention, overcomes the inherent limitation of existing methods that optimize only either lightweightness or feature enhancement. It achieves accurate detection of elongated, low-contrast fine cracks at extremely low computational cost, meeting the practical requirements of real-time UAV edge inspection.

Currently, this study only validates the model's offline performance on a single PC. In future work, we will focus on onboard model quantization, shadow-suppression preprocessing, and dedicated feature branches for ultra-fine cracks. Our goal is to deploy the algorithm on actual UAV hardware for field road inspection, advancing the practical engineering application of UAV-based pavement defect detection.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Code availability statement

The complete code supporting the findings of this study is available from the corresponding author upon request. The authors confirm that any code required to support or reproduce the findings described in this paper will be provided to journal editors, reviewers, and readers without undue restriction or delay upon request.

Conflict of interest

The authors declare no conflicts of interest.

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